C++20

Language

Autocomplete Ready

```
#include <bits/stdc++.h>

using namespace std;

string ltrim(const string &);
string rtrim(const string &);



/*
 * Complete the 'taxiDriver' function below.
 *
 * The function is expected to return a LONG_INTEGER.
 * The function accepts following parameters:
 *  1. LONG_INTEGER_ARRAY pickup
 *  2. LONG_INTEGER_ARRAY drop
 *  3. INTEGER_ARRAY tip
 */

long taxiDriver(vector<long> pickup, vector<long> drop,
{

```

3. Tesla Taxi

A tesla driver knows the pick-up and drop-off locations of people who are requesting taxi services. All the locations are in *km* from the starting point. The starting point is at *0 km.*

For each *km* travelled by a passenger, the driver charges *1* unit of money per passenger. Moreover, as taxi is electric some people are even willing to pay an extra tip if they get to travel in the taxi. At any point of time, the taxi can only accommodate one passenger. Determine the maximum amount the driver can earn.

Example

pickup = [0, 2, 9, 10, 11, 12]

drop = [5, 9, 11, 11, 14, 17]

tip = [1, 2, 3, 2, 2, 1]

The way to earn the most money is by accepting passengers at indices 1, 2 and 5.

- The amount paid by the passenger at index 1: 9 - 2 + 2 = 9
- The amount paid by the passenger at index 2: 11 - 9 + 3 = 5
- The amount paid by the passenger at index 5: 17 - 12 + 1 =6
- The total amount paid by the passengers is 9 + 5 + 6 = 20

Therefore, the return value is 20.

Function Description

Complete the function *taxiDriver* in the editor below. The function must return an integer denoting the maximum amount that can be earned by the driver.

taxiDriver has the following parameter(s):

pickup[pickup[0],...pickup[n-1]]: an array of *n* integers that denote the pickup location of the potential riders

drop[drop[0],...drop[n-1]]: an array of *n* integers that denote the drop-off locations of the potential riders

tip[tip[0],...tip[n-1]]: an array of *n* integers that denote the tips offered by each person if they are accepted for a ride

Constraints

- $0 < |pickup|, |drop|, |tip| \leq 10^4$
- $0 \leq pickup_i, drop_i \leq 10^{12}$
- $pickup_i < drop_i$
- $0 \leq tip \leq 10^5$

▼ Input Format For Custom Testing

The first line contains an integer, *n*, the number of elements in *pickup*.

Each line *i* of the *n* subsequent lines (where $0 \leq i < n$) contains an integer, $pickup_i$.

The next line contains the same integer, *n*, the number of elements in *drop*.

Each line *i* of the *n* subsequent lines (where $0 \leq i < n$) contains an integer, $drop_i$.

The next line contains the same integer, *n*, the number of elements in *tip*.

Each line *i* of the *n* subsequent lines (where $0 \leq i < n$)

Each line i of the n subsequent lines (where $0 \le i < n$) contains an integer, tip_i.

▼ Sample Case 0

Sample Input For Custom Testing

```
STDIN      Function
-----      --------
2        → pickup[] size n = 2
1        → pickup[] = [ 1, 4 ]
4
2        → drop[] size n = 2
5        → drop[] = [ 5, 6 ]
6
2        → tip[] size n = 2
2        → tip[] = [ 2, 5 ]
5
```

Sample Output

```
7
```

Explanation

There are two people, and locations are overlapping so only one of them can be accepted.

If person 1 is picked, the amount made is 5-1+2 = 6

If person 2 is picked, the amount made is 6-4+5 = 7

It is best to pick person 2 and earn 7.

▼ Sample Case 1

Sample Input For Custom Testing

```
STDIN      Function
-----      --------
3        → pickup[] size n =
3
0        → pickup[] = [ 0, 4, 5 ]
4
5
3        → drop[] size n = 3
3        → drop[] = [ 3, 5, 7 ]
5
7
3        → tip[] size n = 2
1        → tip[] = [ 1, 2, 2 ]
2
2
```

Sample Output

```
11
```

Explanation

All three passengers can be accepted because they do not overlap.